

(do not allow more massive so less movement)

3	(a) (i)	$\Delta E_p = mg\Delta h$ $= 0.602 \times 9.8 \times 0.086$ $= 0.51 \text{ J}$ (do not allow $g = 10$, $m = 0.600$ or answer 0.50 J)	C1 A1	[2]
	(ii)	$v^2 = (2gh) = 2 \times 9.8 \times 0.086$ <u>or</u> $(2 \times 0.51)/0.602$ $v = 1.3 \text{ (m s}^{-1}\text{)}$	M1 A0	[1]
	(b)	$2 \times V = 602 \times 1.3$ (allow 600) $V = 390 \text{ m s}^{-1}$	C1 A1	[2]
	(c) (i)	$E_k = \frac{1}{2}mv^2$ $= \frac{1}{2} \times 0.002 \times 390^2$ $= 152 \text{ J or } 153 \text{ J or } 150 \text{ J}$	C1 A1	[2]
	(ii)	E_k not the same/changes <u>or</u> E_k before impact $> E_k$ after/ E_p after so must be inelastic collision (allow 1 mark for 'bullet embeds itself in block' etc.)	M1 A1	[2]